Database Normalization Cheat Sheet

3NF - BCNF - Minimal Cover - Decomposition

Functional Dependencies (FDs)

- FD: $X \to Y$ means: same X implies same Y.
- **LHS**: left-hand side (X); **RHS**: right-hand side (Y).
- **Key**: X^+ = all attributes of the relation.
- **Prime attribute**: any attribute that appears in at least one candidate key.

Normal Forms

BCNF: For every FD $X \to Y$, X must be a superkey.

3NF: For every FD $X \to Y$:

- X is a superkey, **or**
- every attribute in Y is prime (a prime attribute is an attribute that is part of at least one candidate key).

Minimal Cover (Minimal Basis)

To compute a minimal cover for a set of FDs:

1. **Decompose RHS:** ensure each FD has a single attribute on the RHS.
2. **Minimize LHS:** remove extraneous attributes from the LHS.
3. **Remove redundant FDs:** eliminate any FD derivable from the others.

3NF Decomposition Algorithm

1. Compute a minimal cover for the FDs.
2. Create one relation for each FD $X \to A$ in the minimal cover.
3. If no created relation contains a candidate key of the original schema, add one that does.

This algorithm guarantees:

- **Lossless join**
- **Dependency preservation**

Lossless Join Condition

A decomposition of R into R_1 and R_2 is **lossless** if:

$$(R_1 \cap R_2) \to R_1 \quad \text{or} \quad (R_1 \cap R_2) \to R_2.$$

Interpretation:

- $R_1 \cap R_2$ is the set of shared attributes.
- If the shared attributes functionally determine all attributes of R_1 or R_2, then the join cannot invent spurious tuples.

Dependency Preservation

A decomposition preserves dependencies if:

$$(F_1 \cup F_2 \cup \dots)^+ = F^+,$$

i.e., the union of all projected FDs implies all original dependencies.